

CS & IT ENGINEERING



'C' Programming

Control Statements

Lecture No.- 01



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Recap of Previous Lecture



- Exploration of `Printf()`
- Nested `Printf()`
- `Printf()` return value
- Printing Integers, Char, float values
- format specifiers
- Escape sequences

Topics to be Covered



- scanf() function
- Control statements in 'C'
- Conditional Control Statements



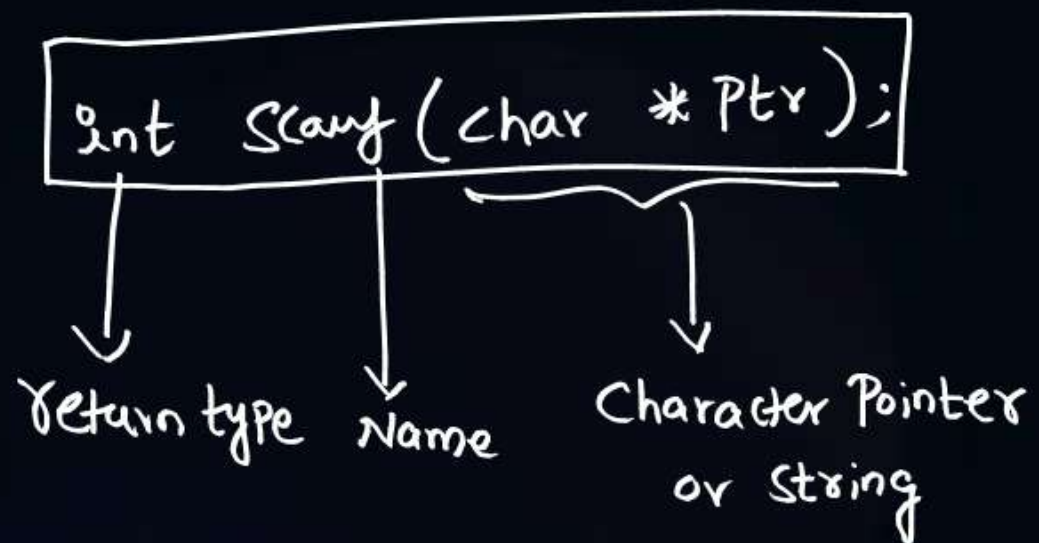


Topic : I/O Functions – scanf()



i/p function – scanf() function

– formatted input function



– scanf() accepts input at runtime [Dynamic i/p], and returns no. of inputs accepted as return value.

Ex: `int a, b;`

`b = scanf("%d", &a);`
`Printf("a = %d, b = %d", a, b);`

i/p: 532 ↵

o/p: a = 532, b = 1



Topic : I/O Functions – scanf()



Syntax:

`scanf("format specifier(s)", &Variable(s));`
↳ "address of"

Ex:

`int a, b, c;`

`scanf("%d %d %d", &a, &b, &c);` i/p: 5 3 7

`printf("a=%d, b=%d, c=%d", a, b, c);`

`a=5, b=3, c=7`

Note: No special symbol except space in b/w format specifiers.

`scanf("%d %d %d", &a, &b, &c);`

↳ Terminator

⇒ It accepts only one input.

`scanf("%d %d %d", &b, &c, &a);`

i/p: 5 3 7

`printf("a=%d, b=%d, c=%d", a, b, c);`

o/p: `a=7, b=5, c=3`



Consider the following C program.

```
#include<stdio.h>
```

```
int main () {
```

```
    int m=10;
```

```
    int n, n1;
```

```
    n=++m;
```

```
    n1=m++;
```

```
    n--;
```

```
    --n1;
```

```
    n-=n1;  $n = n - n1 \Rightarrow 10 - 10 = 0$ 
```

```
    printf("%d", n);
```

```
    return 0;
```

```
}
```

The output of the program is 0

m $\boxed{10 \rightarrow 12}$ n $\boxed{10 \rightarrow 0}$ n1 $\boxed{10}$



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A	B	C	...	Z		a	b	c	...	z
65	66	67	...	90		97	98	99	...	122

*	+	-
42	43	45

What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[]){
    char a = 'P';
    char b = 'x';
    char c = (a & b) + '*';
    char d = (a | b) - '-';
    char e = (a ^ b) + '+';
    printf("%c %c %c\n", c, d, e);
    return 0;
}
```

ASCII encoding for relevant characters is given below

afb $\Rightarrow$ 'P' & 'x' = 80 & 120

$$\begin{array}{r} 80 = 01010000 \\ 120 = 01111000 \\ \hline 01010000 \\ = 80 \end{array}$$

$80 + '*' = 80 + 42 = 122 \Rightarrow$ ASCII character = 'Z'

- a) z K S
b) 122 75 83
c) * - +
d) P x +



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char a = 'P';

char b = 'x';

char c = (a & b) + '*';

char d = (a | b) - '-';

char e = (a ^ b) + '+';

$$a \wedge b = 'P' \wedge 'x' = 80 \wedge 120$$

$$80 = 01010000$$

$$120 = 01111000$$

$$\underline{01111000} = 120$$

$$120 - '-' \Rightarrow 120 - 45 = 75 \Rightarrow \text{ASCII char} = 'K'$$

A	B	C	...	Z		a	b	c	...	z
65	66	67	...	90		97	98	99	...	122

*	+	-
42	43	45

$$a \wedge b = 'P' \wedge 'x' = 80 \wedge 120$$

$$80 = 01010000$$

$$120 = 01111000$$

$$\underline{00101000} = 40$$

$$40 + '+' = 40 + 43 = 83$$

$$\Rightarrow \text{ASCII char} = 'S'$$



Control Statements ✓

- The statement, which can control (change) the order (sequence) of execution of other statements in a program is said to be control statement.

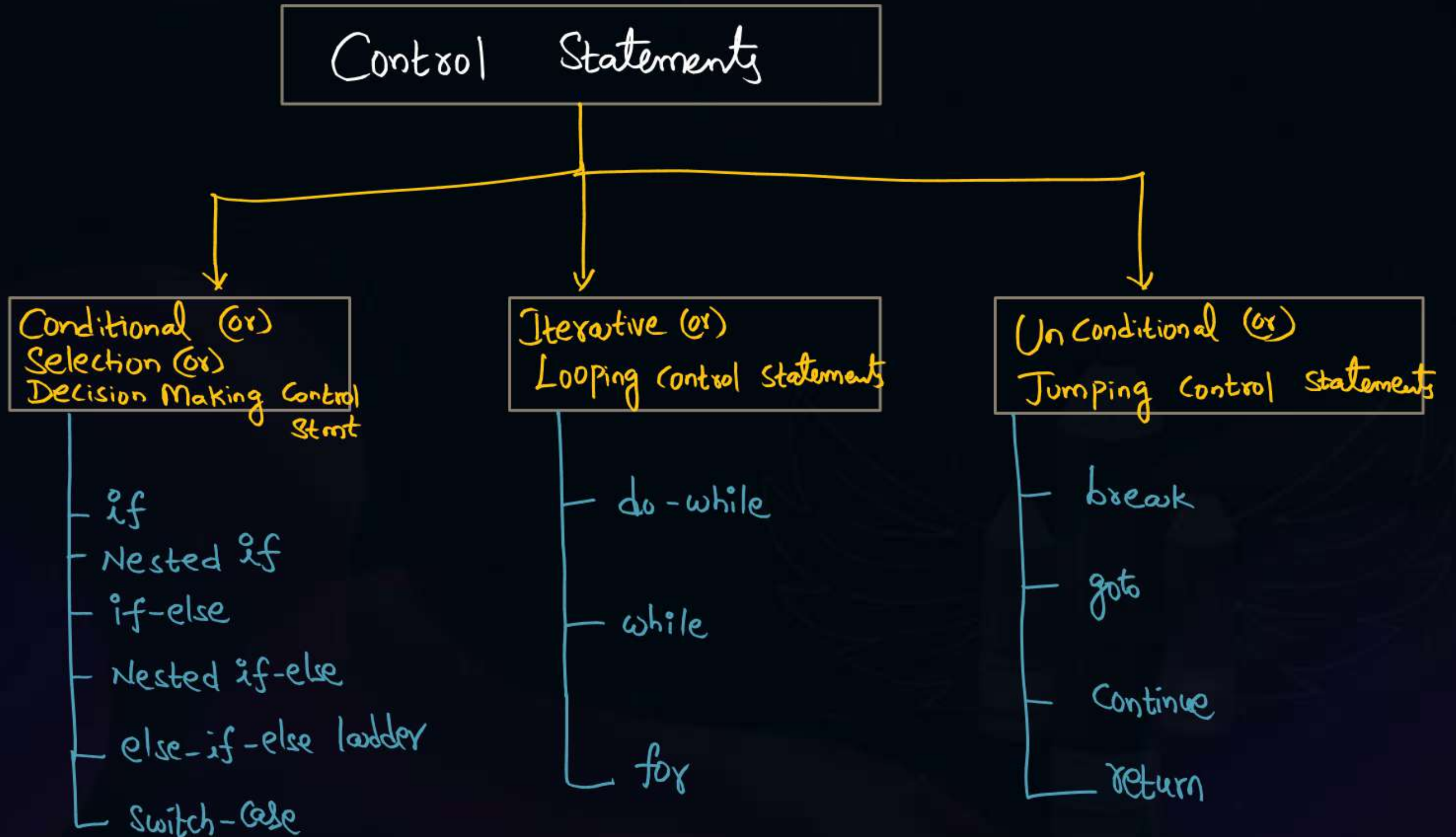
Ex: Until condition is True
keep doing stmt A.

Ex: Only when condition is True
statement

Ex: When condition is True
do statement A
otherwise
do statement B

Ex: When condition 1 is True
do stmt A
when condition 2 is True
do stmt B
when condition 3 is True
do stmt C

Ex: When condition 1 is True
when condition 2 is True
do stmt A
otherwise
do stmt B
otherwise
do stmt C





Topic : Control Statements - 1



if Statement

Syntax:

```
if (Expression)
{
    Statement(s)
}
```

Execute only when
given Expression results
True.

NOTE: When there is only one
Statement with in block,
{ } are optional.

Examples

①
int a = 3, b = 4;
if (a < b)
 printf("a is big");
o/p: a is big

② int x = -3, y = 0
if (y) if(0) → False
 printf("Hello"); X
if (x) if(-3) → Non zero = True
 printf("Hai"); ✓
o/p: Hai

③ int a, b, c;
a = printf("Hai"); a = 3
b = printf(""); b = 0
c = printf(""); c = 1
if (b) if(0) False
 printf("Hello"); X
if (c) if(1) True
 printf("Hey"); ✓
if (a) if(3) True
 printf("Bye"); ✓
o/p: Hai Hey Bye
 space



Topic : Control Statements - 1



④ $\text{if}(\text{Printf}(\text{"Hai"})) \Rightarrow \text{if}(3) \text{ True}$
 $\text{Printf}(\text{" Hello"}); \checkmark$
 $\text{if}(\text{Printf}(\text{" Bye"})) \Rightarrow \text{if}(3) \text{ True}$
 $\text{Printf}(\text{" Hey"}); \checkmark$

o/p: HaiHelloByeHey

⑤ $\text{int } a, b, c;$
 $a = 1, 2, 0; \xrightarrow{\text{ignored}} a = 1$
 $b = 0, 1, 2; \Rightarrow b = 0$
 $c = (1, 2, 0) \quad c = 1 \neq 0$

 $\text{if}(a) \text{ True}$
 $\text{Printf}(\text{" Hello"}); \checkmark$

 $\text{if}(b) \text{ False}$
 $\text{Printf}(\text{" Hai"}); \times$

 $\text{if}(c) \text{ False}$
 $\text{Printf}(\text{" Bye"}); \times$

o/p: Hello



2 mins Summary



⇒ Control stmts (Conditional - if)
⇒ scanf ().

Home work

```
int i=2, j=3, k=0;
```

```
if (i++, j--, --k)
```

```
printf("Hello\n");
```

```
if (i, j, k)
```

```
printf("Hai\n");
```

```
if (--i, ++j, k++)
```

```
printf("Hey\n");
```

```
i++;
```

```
--j;
```

```
k++;
```

```
if (i, j, k)
```

```
printf("Bye");
```

o/p: _____



THANK - YOU